UP504 **Basic Linear Model**

Recall

$$y = m(\underline{x}) + e$$

where $m(\underline{x}) \equiv E[y|\underline{x}] = \int_{-\infty}^{\infty} y f(y|\underline{x})\, dx$

Key properties of the regression error

1. $E[e|\underline{x}] = 0$
2. $E[e] = 0$
3. $E[e\underline{x}] = 0$

skedasis ≡ dispersion

Homoskedasticity vs. heteroskedasticity

a) $E[e^2|\underline{x}] = \sigma^2$

b) $E[e^2|\underline{x}] = \sigma^2(\underline{x})$

skedastic function

conditional variance ($m^2(\underline{x})$)

<u>Old view</u>: Variance of e varies across observations

<u>New view</u>: Conditional variance $\sigma^2(\underline{x})$ depends on $\underline{x}$

$$\sigma^2 = E[e^2] = E[E[e^2|\underline{x}]] = E[\sigma^2(\underline{x})]$$

↳ Standard approach focuses on conditional mean only and assumes that <u>conditional variance is fixed</u> (ie homoskedasticity)

↳ Economist with one foot in bucket with boiling water, other foot in bucket of ice "feels fine on average" → Ignore variance at your peril.

<u>Linear model</u>:

$$y = \underline{x}'\beta + e$$

$$E[e|\underline{x}] = 0$$

$$E[e^2|\underline{x}] = \sigma^2 \text{ or } \sigma^2(\underline{x})$$

↳ <u>Key application</u>: ARCH & GARCH models in finance to model time-varying volatility

AR(n) ARMA(n,q) process

Robert Engle (1982) gets 2003 Nobel Prize

Best linear predictor

$$F(\beta) = E[(\underset{1\times1}{y_i} - \underset{1\times k}{\underline{x}_i'}\underset{k\times1}{\underline{\beta}})^2]$$

[Dropping subscripts in following lines]

$$F(\beta) = E[y^2] - 2\beta' E[\underline{x}y] + \beta' E[x\underline{x}']\beta$$

↳ First order condition:

$$\frac{\partial F(\beta)}{\partial \beta} = -2E[\underline{x}y] + 2E[x\underline{x}']\beta = 0$$

↳ <u>iff $E[x\underline{x}']$ is invertible</u> (non-singular matrix with a positive determinant)

↳ singular matrix ≡ square matrix without an inverse, zero determinant

$$\therefore \quad 2E[\underline{x}y] = 2E[x\underline{x}']\beta$$

$$\therefore \quad \beta = (E[\underline{x}x'])^{-1} E[\underline{x}y]$$

<u>Empirical analog of expected squared error</u> is the <u>sample average squared error</u>.

$$F_n(\beta) = \tfrac{1}{n}\sum(y - \underline{x}'\beta)^2$$

$$= \tfrac{1}{n} SSE_n(\beta)$$

$$SSE_n(\hat{\beta}) = \sum y_i^2 - 2\beta' \sum \underline{x}_i y_i + \beta' \sum \underline{x}_i \underline{x}_i' \beta$$

$$\frac{\partial SSE(\hat{\beta})}{\partial \beta} = 0 = -2\sum \underline{x}_i y_i + 2\sum \underline{x}_i \underline{x}_i' \beta$$

$$\hookrightarrow \hat{\beta} = \left(\sum \underline{x}_i \underline{x}_i'\right)^{-1} \left(\sum \underline{x}_i y_i\right)$$

<u>The model in matrix notation</u>

$$y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \beta_k x_{1k}$$

$$y_1 = \underline{x}_1' \beta + e_1 \qquad \text{ie } \underline{x}_1' = (x_{11}\ x_{12} \ldots x_{1k})$$

$$y_2 = \underline{x}_2 \beta + e_2$$

$$\vdots$$

$$y_n = \underline{x}_n' \beta + e_n$$

since $\underline{x}' = 1 \times k$

$$\hookrightarrow \text{define} \quad \underset{n\times 1}{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \underset{n\times k}{X} = \begin{pmatrix} \underline{x}_1' \\ \underline{x}_2' \\ \vdots \\ \underline{x}_n' \end{pmatrix}, \quad e = \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix}$$

$$\therefore \quad y = X\beta + e$$

with $\sum_1^n \underset{k\times 1}{\underline{x}_i} \underset{1\times k}{\underline{x}_i'} = \underset{k\times n,\ n\times k}{X'X}$ $\quad k \times k$ and

$$\sum \underline{x}_i y = X'y \qquad k\times n \ \ n\times 1 = k\times 1$$

$$\therefore \quad \hat{\beta} = (X'X)^{-1}X'y$$

and $\hat{e} = y - X\hat{\beta}$, $\hat{\sigma}^2 = n^{-1}\hat{e}'\hat{e}$

<u>Matrix review</u>

$$AA^{-1} = A^{-1}A = I_k \quad \text{for a } k \times k \text{ matrix}$$

Matrix A is said to be <u>nonsingular</u> if it is of full rank; ie $\text{rank}(A) = k$

$$A = [a_1 \ a_2 \ \dots \ a_r] \qquad k \times r$$

has rank j iff it has j independent columns

$\therefore$ nonsingularity $\equiv$ full rank (no linearly dependent columns)

$\hookrightarrow \therefore$ invertibility

$\rightarrow$ cf. NONSINGULARITY and MULTICOLLINEARITY

Matrix determinant as measure of matrix volume for as square matrix, representing a rhomboid

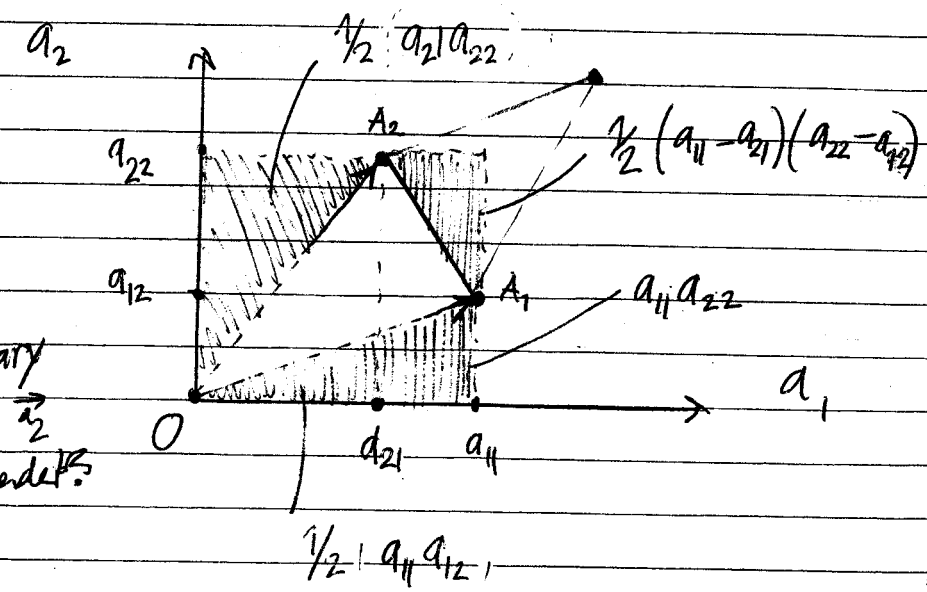
$\det(A) \neq 0$ iff A is non-singular

Consider: $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

det $\equiv$ sum of all signed elementary products

$\therefore \det(A) = a_{11}a_{22} - a_{12}a_{21}$

$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A) = \frac{1}{\det A}\begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$ $\det(A^{-1}) = \frac{1}{\det(A)}$



What happens if 2 elementary vectors $\vec{a}_1$ and $\vec{a}_2$ are linearly dependent?

6 $\therefore \cancel{a_{11}a_{22}} - \cancel{\frac{1}{2}a_{11}a_{12}} - \cancel{\frac{1}{2}a_{21}a_{22}} - \cancel{\frac{1}{2}a_{11}a_{22}} + \cancel{\frac{1}{2}a_{11}a_{12}} + \cancel{\frac{1}{2}a_{21}a_{22}} - \cancel{\frac{1}{2}a_{12}a_{21}}$

$= \frac{1}{2}a_{11}a_{22} - \frac{1}{2}a_{12}a_{21}$

ACS Tables

Margins of error → permit construction of 90% confidence bands

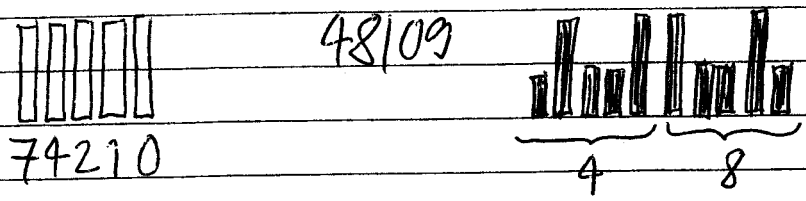
$$s.e = (MOE_{ACS}) / 1.645$$

Eg. Test difference in population means

$$\left| \frac{\mu_1 - \mu_2}{\sqrt{s_1^2 + s_2^2}} \right| > z_{confidence}$$

∴ Standard error = MOE / 2 → Margin = s.e. × z-score

PostNet ZIP-codes



* HEXadecimal colours (RGB range 0–255)

00 through FF with 0–9, A, B, C, D, E, F

0×16^1, 0×16^0

eg. $201 \rightarrow 12 \times 16 + 9 \times 1 = 69$

UP504 <u>OLS continued</u>

<u>Excursion</u> : $(X'X)^{-1}$ as a transforming space

$$\hat{\beta} = (X'X)^{-1}X'y$$

↳ Now assume that $(X'X)^{-1} = I$, this would imply that:

(i) off-diagonal elements 0; ie data is uncorrelated

(ii) diagonal with ones; ie sum of squared variable $= 1$ which implies $\mu_x = 0$ and $\sigma_x = 1/N$ $\quad (X'X)^{-1} = I \rightarrow X'X = I$ because $\therefore X' = X^{-1}$

↳ Then $\hat{\beta} = IX'y = \boxed{X'y} \equiv X^{-1}y$ NAIVE REGRESSION COEFF.

$\therefore$ The rotating matrix thus compresses, expands and rotates the naive regression coefficients

* ↳ $(X'X)^{-1}$ as <u>corrective lens</u> based on <u>scale and correlations</u> of original regressors.

$\therefore$ Similar scaling interpretation of $s^2(X'X)^{-1}$, thus linking the variance of the residuals to the standard errors of the regression coefficients.

<u>Recall</u>: $\hat{\beta}$ is also a random variable with its own $\mu_{\hat{\beta}}$ and $\sigma^2_{\hat{\beta}}$

↳ Thus it has a sampling distribution

$$E(\hat{\beta} \mid X) = \beta \quad \text{and} \quad E(\hat{\beta}) = \beta$$

$$\downarrow$$

$$E(\hat{\beta} \mid X) = E[(X'X)^{-1}X'y \mid X] =$$
$$= (X'X)^{-1}X'E(y \mid X) = (X'X)^{-1}X'X\beta$$
$$= \beta$$

What about the variance of $\hat{\beta}$, $Var(\hat{\beta})$?

$$Var(e) = E(\underset{n\times 1 \;\; 1\times n}{ee'}) = \underset{1\times 1}{\sigma^2}\underset{n\times n}{I} \quad \text{and}$$
$$\hat{\beta} - \beta = (X'X)^{-1}X'e$$

$$V(\hat{\beta}) = E[(\hat{\beta}-\beta)(\hat{\beta}-\beta)'] = E[(X'X)^{-1}X'ee'X(X'X)^{-1}]$$
$$= (X'X)^{-1}X'E(ee')X(X'X)^{-1}$$
$$= (X'X)^{-1}X'\sigma^2 I X(X'X)^{-1}$$
$$= \sigma^2(X'X)^{-1}X'X(X'X)^{-1} = \sigma^2(X'X)^{-1}$$

Estimate of error variance σ^2:

OLS does not provide a direct estimate of σ^2. If e was actually observable, then

$$\sigma^2 = \frac{1}{n}\sum e_i^2 \qquad \text{(why? mean zero...)}$$

But best we can do is actually

$$\hat{\sigma}^2 = \frac{1}{n}\sum \hat{e}_i^2$$

However each squared residual can be shown to have an expectation that is smaller than σ^2 such that

$$E[\hat{\sigma}^2] < \sigma^2 \quad \therefore \hat{\sigma}^2 \text{ \underline{always biased downwards}}$$

We can show that:

$$E[\hat{\sigma}^2] = \frac{n-k}{n}\sigma^2$$

$\therefore$ bias takes "scale form" (depends on ratio of coefficients to sample size k/n), thus unbiased estimator via re-scaling

$$s^2 \equiv \frac{1}{n-k}\sum^{n} \hat{e}_i^2 = \frac{n}{n-k}\hat{\sigma}^2$$

* NB: while s^2 is an unbiased estimate of σ^2, s does <u>not</u> provide unbiased estimate of σ!

⊛ inner product:

$$a^T b = b^T a = \sum_i^n a_i b_i$$

Basic Matrix Algebra revisited

$$\underline{a}_i = \begin{pmatrix} a_{1i} \\ a_{2i} \\ \vdots \\ a_{ni} \end{pmatrix} \longrightarrow \underset{n \times k}{A} = [\underline{a}_1 \ \underline{a}_2 \ \dots \ \underline{a}_k]$$

$$= \begin{bmatrix} \underline{\alpha}_1 \\ \underline{\alpha}_2 \\ \vdots \\ \underline{\alpha}_n \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & & \vdots \\ \vdots & & \ddots & \vdots \\ a_{n1} & \cdots & & a_{nk} \end{bmatrix}$$

column vector

$$a_i^T =$$

$$\underline{\alpha}_i = (\alpha_{j1} \ \alpha_{j2} \ \dots \ \alpha_{jk})$$

row vector

→ special vector ι: ⊛
with $\iota^T a = \sum a_i = n\bar{a}$
summation of vector elements
$\therefore \iota^T \iota = n$

Recall the OLS model in matrix notation:

⊛ (1)

$$\left.\begin{aligned} y_1 &= \underline{x}_1' \underline{\beta} + e_1 \\ y_2 &= \underline{x}_2' \underline{\beta} + e_2 \\ &\vdots \\ y_n &= \underline{x}_n' \underline{\beta} + e_n \end{aligned}\right\}$$

with $\underline{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$, $X = \begin{pmatrix} \underline{x}_1' \\ \underline{x}_2' \\ \vdots \\ \underline{x}_n' \end{pmatrix}$, $\underline{e} = \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix}$

(12) ⊛ (2) $\underline{y} = X\underline{\beta} + \underline{e}$

Note: 3 different ways to write linear model

⊛ (3) $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots \beta_{k-1} x_{k-1,i} + e_i$

Important notational equivalents:

$$\sum_{i=1}^{n} \underline{x}_i \underline{x}_i' = X'X$$

$$\sum_{i=1}^{n} \underline{x}_i y_i = X'y$$

since $\underset{n \times k}{X} = \begin{bmatrix} \underline{x}_1' \\ \underline{x}_2' \\ \vdots \\ \underline{x}_n' \end{bmatrix}$ and $\underset{k \times n}{X'} = [\underline{x}_1 \ \underline{x}_2 \ \dots \ \underline{x}_n]$

$$\therefore \ X'X = \underset{k}{[\underline{x}_1 \ \underline{x}_2 \ \dots \ \underline{x}_n]} \begin{bmatrix} \underline{x}_1' \\ \underline{x}_2' \\ \vdots \\ \underline{x}_n' \end{bmatrix} = \underline{x}_1 \underline{x}_1' + \underline{x}_2 \underline{x}_2' + \dots + \underline{x}_n \underline{x}_n'$$

$$k \begin{bmatrix} \Sigma x_1^2 & \Sigma x_1 x_2 & \cdots & \Sigma x_1 x_k \\ \Sigma x_2 x_1 & \ddots & & \\ \vdots & & & \\ \Sigma x_k x_1 & \cdots & & \Sigma x_k^2 \end{bmatrix} = \sum_{i=1}^{n} \underline{x}_i \underline{x}_i'$$

Also; sum of squares $\Sigma x_i^2 = \underline{x}'\underline{x}$ (for x as column vectors)

Centering of variables (deviations from mean)

Eg. $\underline{x}$ and its mean $\bar{x} = \frac{1}{n}\sum x_i$

Let $\underline{z} \equiv \underline{x} - \bar{x}\underline{\iota}$

ie. inner product is zero

We can now show that the centered variable $\underline{z}$ is orthogonal to the constant $\underline{\iota}$

$$\underline{\iota}'\underline{z} = \underline{\iota}'(\underline{x} - \bar{x}\underline{\iota}) = \underline{\iota}'\underline{x} - \bar{x}\underline{\iota}'\underline{\iota} = n\bar{x} - \bar{x}n = 0$$

Because centering leads to a variable that is orthogonal to $\underline{\iota}$, it can be performed by the orthogonal projection matrix M_ι

$$M_\iota \equiv I_n - P_\iota = I - \underline{\iota}(\underline{\iota}'\underline{\iota})^{-1}\underline{\iota}' = I - \underline{\iota}\underline{\iota}'/n$$

↳ Proof:

$$M_\iota \underline{x} = (I - P_\iota)\underline{x} = \underline{x} - \underline{\iota}(\underline{\iota}'\underline{\iota})^{-1}\overbrace{\underline{\iota}'\underline{x}}^{n\bar{x}} =$$

$$= \underline{x} - \bar{x}\underline{\iota} = \underline{z}$$

From
the Goodness of Fit to Specification
<u>Application to regression analysis</u> R^2 (centered R^2)

Recall that:

$$TSS = ESS + RSS$$

$$\sum^n (y_i - \bar{y})^2 = \sum^n (\hat{y}_i - \bar{y})^2 + \sum (y_i - \hat{y}_i)^2$$
$$\sum (y_i - \bar{y})^2 + \sum \hat{e}_i^2$$

Plus: $y = P_X y + M_X y \quad \therefore \hat{y} = P_X y$

↳ $y - \bar{y} \equiv$ centered variable by multiplying by M_i

$\therefore M_i y = y - \bar{y}$ and recall that $y'y = \sum y^2$

$$\therefore \boxed{TSS = y' M_i y}$$

Similarly:

$$\boxed{\begin{aligned} ESS &= y' M_i P_X y \\ RSS &= y' M_X y \end{aligned}}$$

and

And thus: $R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS}$

<u>NB</u>: Adjusted $R^2 = 1 - (1 - R^2)\frac{n-1}{n-k}$ with $k-1$ regressors

↳ $\bar{R}^2$ increases only if t-stat of new variable is greater $|1|$

∴ increase in $\bar{R}^2$ generally not indicative of improved specification

→ Applied econometricians increasingly pay less and less attention to specification optimisation via R^2

Even a poorly specified equation can achieve a high R^2.

UP 504 Geometry of OLS

$\hat{\beta}$ is a random variable with its own $\mu_{\hat{\beta}}$ and $\sigma^2_{\hat{\beta}}$

$$\hat{\beta} = (X'X)^{-1}X'y = (X'X)^{-1}X'(X\beta + e)$$
$$= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'e$$
$$= \beta + (X'X)^{-1}X'e$$

$\downarrow$

$E(e|X) = 0$

$\hookrightarrow \hat{y} = X\hat{\beta} = X(X'X)^{-1}X'y = P_X y$

Orthogonal projections

hat matrix

$P_X = X(X'X)^{-1}X'$

$M_X = I - P_X = I - X(X'X)^{-1}X'$

$P_X P_X = P_X$

idempotent matrix

$P_X + M_X = I$

complementary projection matrices

$\hookrightarrow$ orthogonal complements which restore the original vector $y = P_X y + M_X y$

$\therefore\ P_X M_X = 0$ since $P_X(I - P_X) = P_X - P_X P_X$

$$= P_X - P_X = 0$$

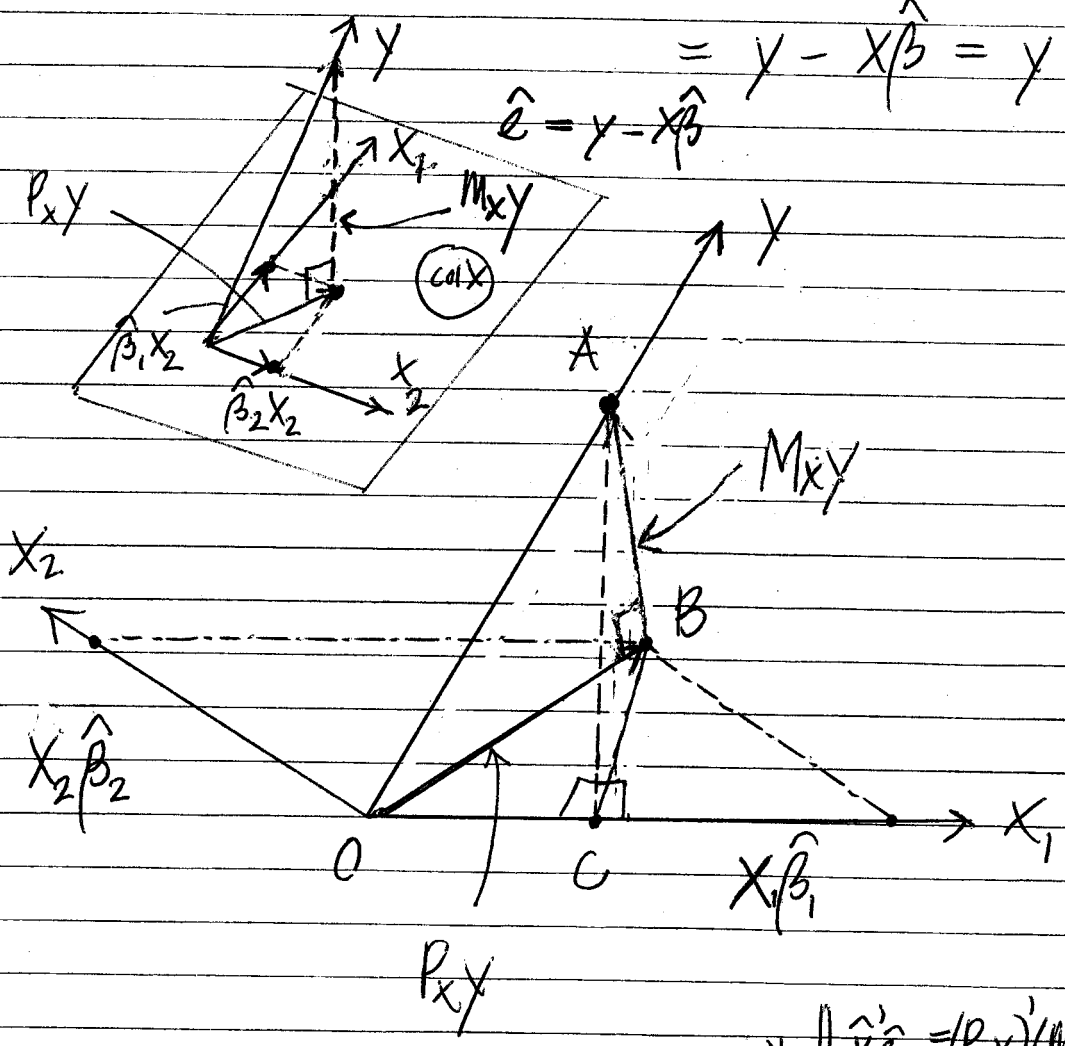
$$M_X X = 0$$

$$X\hat{\beta} = P_X y = \hat{y}$$

$$M_X y = (I - P_X)y = y - P_X y$$

$$= y - X\hat{\beta} = y - \hat{y} = \hat{e}$$

$$\hat{e} = y - X\hat{\beta}$$



$$* \left\| \ \hat{y}'\hat{e} = (P_X y)'(M_X y) = y' P_X M_X y = 0 \right.$$

$$y = P_X y + M_X y \equiv \text{orthogonal decomposition}$$

$$\hat{y} + \hat{e}$$

$\hookrightarrow \sigma^2 = n^{-1}\hat{e}'\hat{e} = n^{-1} y' M M y = n^{-1} y' M y$

OLS as a more general estimator:

$$\hat{\beta} = (Z'Z)Z'y \qquad \text{eg. } \hat{\beta}_{2SLS} \equiv \beta_{IV} \text{ if}$$

just-identified

$$\begin{tikzcd} x \arrow[r] & y \\ e \arrow[ur] & \end{tikzcd} \qquad \text{but what if} \qquad \begin{tikzcd} \overset{\text{ability}}{z} \arrow[r] & \overset{\text{schooling}}{x} \arrow[r] & \overset{\text{wage}}{y} \\ & e \arrow[u] \arrow[ur, dashed] & \end{tikzcd}$$

$$\hat{\beta}_{2SLS} = \left[X'Z(Z'Z)^{-1}Z'X\right]^{-1}\left[X'Z(Z'Z)^{-1}Z'y\right]$$

with first stage:

$$\hat{X} = Z(Z'Z)^{-1}Z'X \quad \text{ie} \quad \hat{\beta}_{1stage} = (Z'Z)^{-1}Z'X$$

$$\hat{\beta}_{IV} = (Z'X)^{-1}X'y$$

[see p. 24 for FWL theorem]

GDP and GDI (from NIPA)

<u>GDI</u> (2003)

- Compensation of employees (56%)
- Taxes on production and imports (TOPI)
  - less subsidies (7%)
- Net operating surplus (25%)
- Consumption of fixed capital (CFC) (12%)

(+ statistical discrepancy ≡ GDP)
~ 0.2%

TOPI ≡ sales taxes, property taxes, custom duties

NOS ≡ corporate profits, net earnings of unincorporated businesses, govt surplus eg. FNMA, Fed profits

CFC ≡ measures depreciation, capex, interest

Measuring price changes:

* CPI : fixed basket of goods over time
* GDP deflator : change in nominal GDP over time

$$= \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

↳ main difference to CPI is that weights of basket are not fixed

2 types of fixed-weight indices (BLS)

1) Paasche index : $\frac{\sum p_{it} q_{it}}{\sum p_{i0} q_{it}}$, understates inflation

2) Laspeyres index : $\frac{\sum p_{it} q_{i0}}{\sum p_{i0} q_{i0}}$, overstates inflation

↳ CPI-W, CPI-U until recently as Laspeyres indices. Now a Fisher ideal index which is geometric mean between P + L

$$\left(\prod_{i=1}^{n} a_i\right)^{1/n} \quad \text{vs} \quad \frac{1}{n}\sum_{i=1}^{n} a_i$$

Transformation of variables (Nonlinearity)

Central assumption of classical linear regression model

* dep. var as linear function of ind. vars
* β is constant

↳ (i) Wrong regressors (inclusion or omission)

(ii) Nonlinearity

(iii) changing parameters (β varies with time or across observations (Rand. coeff. models))

① Transforming single independent variables

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + e$$

↳ let $z = x^2$ to preserve linearity in parameters

② Transform entire equation

Highlight CRS, IRS

Cobb-Douglas production function

$$Y = AK^{\alpha}L^{\beta}e \quad \text{with CES } \alpha = 1-\beta$$

→ log linearisation

$$\ln y = \ln A + (1-\beta)\ln K + \beta \ln L + \ln e \quad (22)$$

Constant returns to scale

$$F(aK, aL) = A(aK)^{\alpha}(aL)^{1-\alpha} = Aa^{\alpha}K^{\alpha}a^{1-\alpha}L^{1-\alpha}$$
$$= a(AK^{\alpha}L^{1-\alpha})$$

What if $\alpha + \beta > 1$?

Elasticity (of substitution):

$$\frac{dY}{Y} \cdot \frac{X}{dX} = \frac{dY/dX}{Y/X} \equiv \frac{\text{marginal product}}{\text{average product}}$$

$$\frac{dY}{dL} = \beta AK^{\alpha}L^{\beta-1} = \beta \frac{Y}{L}$$

$$\therefore \frac{dY}{dL} \Big/ \frac{Y}{L} = \beta \frac{Y}{L} \cdot \frac{L}{Y} = \beta \quad \left[\text{constant factor proportion}\right]$$

Cobb-Douglas derived from CES $Y = A[\alpha K^{\rho} + (1-\alpha)L^{\rho}]^{1/\rho}$

$\hookrightarrow \rho = 0$ yields $Y = AK^{\alpha}L^{1-\alpha}$ from

$$\ln(Y) = \ln(A) + 1/\rho \cdot \ln[\alpha K^{\rho} + (1-\alpha)L^{\rho}]$$

L'Hôpital's rule: $\lim_{\rho \to 0} \ln(Y) = \ln(A) + \alpha \ln(K) + (1-\alpha)\ln(L)$

$$\therefore Y = AK^{\alpha}L^{1-\alpha}$$

Frisch-Waugh-Lovell Theorem (FWL)

If $Y = X_1\beta_1 + X_2\beta_2 + e$

where X_1 and X_2 are $n \times k_1$ and $n \times k_2$ matrices, and where β_1 and β_2 are conformable, then the estimate of β_2 will be the same as the estimate from an auxilliary regression of the form

$$M_{X_1} Y = M_{X_1} X_2 \beta_2 + v \qquad (\text{where } v = M_{X_1} e)$$

where M_X projects onto the orthogonal complement of the image of the projection matrix $X_1(X_1'X_1)^{-1}X_1'$

Equivalently, M_{X_1} projects onto the orthogonal complement of the column space of X_1. Specifically,

$$M_{X_1} = I - X_1(X_1'X_1)^{-1}X_1'.$$

Proof: Let $M_{X_1} \equiv M_1$ for notational convenience

$$Y = X_1\beta_1 + X_2\beta_2 + e$$

$$M_1 Y = \underbrace{\cancel{M_1 X_1 \beta_1}}_{=0,\ \text{since } M_1 X_1 = 0} + M_1 X_2 \beta_2 + v \qquad (v = M_1 e)$$

24 Then: recall that $Y = P_X Y + M_X Y = X_1\hat{\beta}_1 + X_2\hat{\beta} + M_X Y$

multiply previous equation by $X_2' M_1$ such that

$$X_2' M_1 Y = \overset{①}{X_2' M_1 X_1 \hat{\beta}_1} + \overset{②}{X_2' M_1 X_2 \hat{\beta}_2} + \overset{③}{X_2' M_1 M_X Y}$$

make use of the following properties:

1. $M_1 X_1 = 0 \therefore X_2' M_1 X_1 \hat{\beta}_1 = 0$ in ①
2. $M_1 M_X = M_X$ because $M_X < M_1$

$\therefore X_2' M_1 M_X Y = X_2' M_X Y = X_2' \hat{e} = 0!$ in ③

Thus terms ① and ③ disappear in the above equation and we have

$$X_2' M_1 Y = X_2' M_1 X_2 \hat{\beta}_2$$, solving for $\hat{\beta}_2$ this yields

$$\boxed{\hat{\beta}_2 = (X_2' M_1 X_2)^{-1} X_2' M_1 Y}$$

$\therefore$ Due to the FWL theorem, we can estimate β_2 directly using orthogonal projections, without having to estimate β_1.

By extension: $Var(\hat{\beta}_2) = \sigma^2 (X_2' M_1 X_2)^{-1}$

UP 504 <u>Specification issues</u>

<u>Overspecification</u>:

* Variables that do not belong to DGP are mistakenly included in DGP

↳ overspecification is <u>not</u> misspecification

$$y = X\beta + Z\gamma + e \quad \text{with } e \sim iid(0, \sigma^2 I)$$

but actual DGP is $y = X\beta_0 + e$ with $(0, \sigma_0^2 I)$

↳ β is not biased since overspecified model is a special case with

$$\beta = \beta_0 \text{ and } \gamma = 0 \quad (\text{plus } \sigma^2 = \sigma_0^2)$$

<u>Underspecification</u>:

* Important explanatory variables are omitted and we are estimating

$$y = X\beta + e \quad \text{with } e \sim iid \sim (0, \sigma^2 I)$$

but true DGP is given by

$$y = X\beta_0 + Z\gamma_0 + e \quad \text{with } e \sim iid(0, \sigma_0^2 I)$$

↳ <u>Omitted variable bias</u>:

$$\begin{aligned} E(\hat{\beta}) &= E[(X'X)^{-1}X'(X\beta_0 + Z\gamma_0 + e)] \\ &= \beta_0 + (X'X)^{-1}X'Z\gamma_0 + E[(X'X)^{-1}X'e] \\ &= \beta_0 + \underbrace{(X'X)^{-1}X'Z\gamma_0}_{\text{bias}} \end{aligned}$$

depends on 1. $X'Z = 0$ (orthogonal regressors)

2. $\gamma = 0$ (model in fact <u>not</u> misspecified)

* <u>Overspecification</u>: unbiased, (weakly) consistent, normally inefficient estimates

* <u>Underspecification</u>: biased, consistent linear estimator with efficiency that depends on specific sample.

Heteroskedasticity (as misspecification)

New DGP with $E(e_i^2 | x_i) = \sigma^2(x_i) = \sigma_i^2$

$$y_i = X\beta + e_i \quad \text{with} \quad e_i \sim id(0, \Omega)$$

$$\therefore \Omega \equiv diag(\sigma_1^2, \sigma_2^2, \dots \sigma_n^2) \qquad E[ee']$$

OLS is still unbiased when there is heteroskedasticity because $E[\beta]$ only depends on first moments and $E[e] = 0$

↳ recall: $\hat{\beta} - \beta = (X'X)^{-1} E[X'e]$

$\therefore$ does not depend on higher moments

$$V(\hat{\beta}) = E[(\hat{\beta}-\beta)(\hat{\beta}-\beta)'] =$$
$$= E[(X'X)^{-1}X'ee'X(X'X)^{-1}] =$$
$$= (X'X)^{-1}X'E[ee']X(X'X)^{-1} =$$
$$= (X'X)^{-1}(X'\Omega X)(X'X)^{-1}$$

$\therefore$ GLS estimator: $\hat{\beta} = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}y$

Excursion: Greek Alphabet

Αα	alpha
Ββ	beta
Γγ	gamma
Δδ	delta
Εε	epsilon
Ζζ	zeta
Ηη	eta
Θθ	theta
Ιι	iota
Κκ	kappa
Λλ	lambda
Μμ	mu
Νν	nu
Ξξ	xi
Οο	omicron
Ππ	pi
Ρρ	rho
Σσ	sigma
Ττ	tau
Υυ	upsilon
Φφ	phi
Χχ	chi
Ψψ	psi
Ωω	omega

↳ key challenge when implementing GLS;

Ω is not observable $\therefore$ we must estimate $\hat{\Omega}$.

Instrumental variables

OLS as a general estimator of the form

$$\tilde{\beta}_W = (W'W)^{-1} W'y \qquad \text{eg. } \hat{\beta}_{IV} \equiv \hat{\beta}_{2SLS}$$

with just-identification
(ie number of parameters is equal to # of instruments)

2 specific cases that can interfere with our key OLS assumptions; ie correlation b/w error and regressors

* ENDOGENEITY * $\rightarrow E(X'e) \neq 0$

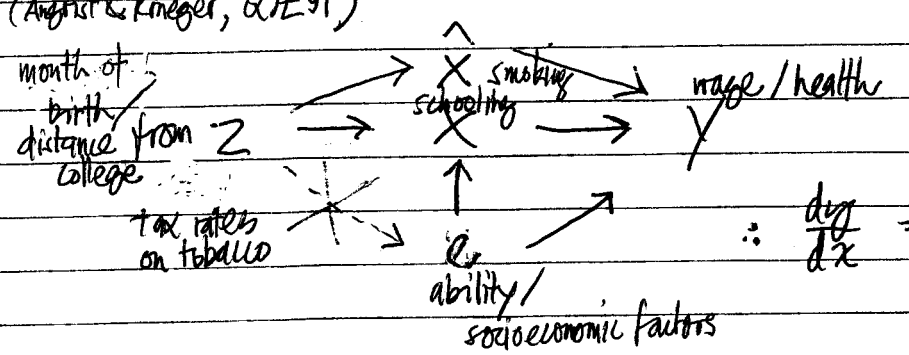
(i) Measurement error

(ii) Endogeneity due to simultaneous equations
eg. demand & supply:

$$q_i = -\beta_1 p_i + e_{1i}$$
$$q_i = \beta_2 p_i + e_{2i}$$

$\boxed{\therefore \hat{\beta} \text{ will be biased and inconsistent!}}$

(Angrist & Krueger, QJE 91)



$$\therefore \frac{dy}{dx} = \beta + \frac{de}{dx}$$

$$\hat{\beta}_{IV} = (Z'X)^{-1} Z'y$$

If Z is a matrix of k instruments, such that

$$y = P_Z X\beta \qquad X = [\,X_1 \;\; X_2\,]$$

ENDOGENOUS REGRESSOR

$$\text{where } P_Z = Z(Z'Z)^{-1}Z' \qquad Z = [\,X_1 \;\; Z_2\,]$$

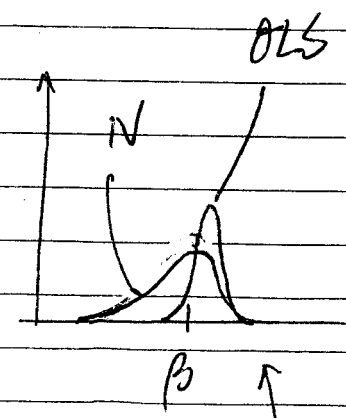
instruments

NB. X_1 instruments for itself

$\therefore$ 2SLS estimator is IV estimator in just-identified case:

$$\hat{\beta}_{2SLS} = (X'P_Z X)^{-1} X'P_Z y$$

$$= [\underbrace{X'Z(Z'Z)^{-1}Z'X}_{\hat{X}'}]^{-1} \underbrace{X'Z(Z'Z)^{-1}Z'}_{\hat{X}'} y$$



with 1. stage regression:

$$\hat{X} = Z(Z'Z)^{-1}Z'X =$$

$$= Z\hat{\beta}_{1stage} \quad \text{where } \hat{\beta}_{1stage} = (Z'Z)^{-1}Z'X$$

(32) * ↳ IV estimator is also biased (but generally less so than OLS) and <u>consistent</u>! (vs. inconsistent in OLS)

Measurement error:

The regressors X^* are not observed and only measured with error u (indep. of X and y)

$$X = X^* + u$$

The model:

$$y = X^*\beta + e$$

$$y = (X-u)\beta + e$$

$$y = X\beta + \underbrace{e - u\beta}$$

$$= X\beta + v$$

↳ $E[\hat{\beta}] = E[(X'X)^{-1}X'y]$

$$= E[(X'X)^{-1}X'(X\beta + v)] =$$

$$= \beta + E[(X'X)^{-1}X'v]$$

↳ determine size of bias from $E[X'v]$:

$$E[X'v] = E[(X^* + u)'(e - u\beta)]$$

$$= E[X^{*\prime}e + u'e - X^{*\prime}u\beta - u'u\beta]$$

$$= -E[u'u]\beta$$

$-E[u'u]\beta \neq 0$ if $\beta \neq 0$ and $E[u'u] \neq 0$

$\hat{\beta} \xrightarrow{p} \beta^*$ (probability limit of $\hat{\beta}$)

$\therefore \ \beta^* = \beta - E[(X'X)^{-1}u'u]\beta$

2SLS Computation

1) Regress X on Z ie $\hat{\Gamma} = (Z'Z)^{-1} Z'X$ and $\hat{X} = Z\hat{\Gamma}$

2) Regress y on $\hat{X}$ ie $\hat{\beta} = (\hat{X}'\hat{X})^{-1} \hat{X}'y$

Scrutinize $\hat{X}$; recalling $X = [X_1 \; X_2]$

$$Z = [X_1 \; Z_2]$$

$$\begin{aligned}
\hat{X} &= [\hat{X}_1, \hat{X}_2] \\
&= [P_Z X_1, P_Z X_2] \\
&= [X_1, P_Z X_2] \\
&= [X_1, \hat{X}_2]
\end{aligned}$$

$\therefore$ only endogenous variables are replaced with their fitted values

$$\hookrightarrow \quad \hat{X}_2 = Z_1 \hat{\Gamma}_{12} + Z_2 \hat{\Gamma}_{22}$$